

## The *in vitro* antifungal activity of 30 Chinese herb extracts to *Saprolegnia* sp.

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### Summary

Chinese traditional medicines have been used for several thousands of years in Asian countries, not only in humans but also in many animal species. These compounds prevent and control different types of diseases including internal diseases as well as some infectious diseases, where the aetiological agent is viral, bacterial, parasitic or mycotic. *Rhizoma coptidis* is believed to inhibit *Shigella dysenteriae* and that *Radix isatidis* can prevent flu caused by the influenza virus. It is thus hypothesized that some of these traditional herbal compounds will have anti-fungal activity. Saprolegniosis is a disease common in fish and their eggs in both fresh and brackish water; a newer, safer medication against *Saprolegnia* is needed after the prohibition in many countries of the extremely effective fungicide, malachite green. In the present study an attempt is made to identify herbal compounds that have anti-*Saprolegnia* activity. A strain of *Saprolegnia*, CCF1301, was isolated from the skin of infected grass carp (*Ctenopharyngodon idella*) and identified as *Saprolegnia ferax* by the 26S rDNA D1/D2 region and ITS region. This strain was used to evaluate the antifungal activity of thirty Chinese traditional herbal medicine extracts, and a modified dish dilution method was developed for the evaluation. *Saprolegnia*-infected rapeseeds with visible hyphae were transplanted onto prepared PDA plates containing 2 g L<sup>-1</sup> herbal plant extracts and incubated at 20°C for 48 h. Each herbal plant species was tested in triplicate. Those herbal plant extracts that showed negative mycelium presence at 2 g L<sup>-1</sup> were further tested for minimum inhibitory concentration (MIC) evaluation. The results showed that *Syzygium aromaticum*, *Magnolia officinalis*, *Melaphis chinensis*, *Euphorbia fischeriana* Steud, and *Sophora flavescens* exhibited enhanced growth inhibition at 2 g L<sup>-1</sup> and MIC values of 500, 62.5, 250, 62.5, 250 mg L<sup>-1</sup> concentrations, respectively. It was obvious that *Magnolia officinalis* and *Euphorbia fischeriana* Steud exhibited the best antifungal activity. Since there is a high natural toxicity in *Euphorbia fischeriana* Steud, its applicability as the main ingredient in an aquaculture therapeutic formulation requires further research. Thus, *Magnolia officinalis* would appear to be the more valuable antifungal herbal species with which to pursue further research.

### Introduction

Saprolegniosis caused by *Saprolegnia* sp. is the most common fungal disease affecting fish and their eggs in fresh and brackish water (Eli et al., 2011). This disease causes serious losses in fish farms and hatcheries and is a key disease in farmed and wild fish in many countries (Ke et al., 2009; Karouei, 2012; Albert et al., 2013; Vega-Ramírez et al., 2013). The hyphae grow through the epidermis and dermis into the musculature and usually cause dermatitis and myositis (Husein and Hatai, 2002). Infected fish show darkened coloration, listlessness, tend to be off their feed and are covered by white, cotton-like hyphae. Extensive superficial infections and deep focal infections can cause loss of osmoregulation via necrosis and hemorrhaging of the skin, leading to death in the severely affected fish (Roberts, 2009).

Malachite green was the most effective therapeutant to control and prevent *Saprolegnia* zoospore production and cyst germination. The use of malachite green has, however, been prohibited to use in fish for human consumption due to its carcinogenic and toxicological effects (The Ministry of Agriculture of the People's Republic of China's, 2002). This ban has necessitated the search for acceptable safe/efficient alternatives. Although sodium chloride and formalin are currently used in fish farms to prevent saprolegniosis, the huge volumes required in comparison to other chemicals makes it impractical for use in large ponds. Traditional Chinese herbal medicines have been used to prevent diseases in humans and animals for thousands of years in China, Japan, Korea and many other Asian countries (Chanu et al., 2012). Currently, many natural plant compounds have been shown to be effective in preventing and controlling fish diseases (Wang and Cui, 2004); nevertheless, research on their antifungal properties has rarely been reported. Furthermore, there is limited or no information as to which herbal compounds are effective against *Saprolegnia* sp. or to the active ingredient(s) in these compounds. We thus have searched to identify effective antifungal therapeutants from thirty naturally-occurring plant species using an isolate of *Saprolegnia* sp. obtained from a natural infection in cultured grass carp (*Ctenopharyngodon idella*). Anticipated is that this will provide a basis to identify the active compound(s) associated with the antifungal activity of the herbal extract, for potential use as an

alternative therapeutant to control saprolegniosis in cultured fish.

## Materials and methods

### Fungal isolation

Five fish (11.4–15.7 g) with visible signs of saprolegniosis symptoms appearing on the back and tail were sampled from the teaching fish farm of Sichuan Agricultural University (SAU). Fish were euthanized by administering a short, sharp blow to the head. The diseased fish were immersed in 75% ethyl alcohol for 2–3 s and washed several times with sterile double distilled water (dH<sub>2</sub>O). Hyphae were carefully removed onto plates of potato dextrose agar (PDA) with 200 mg L<sup>-1</sup> streptomycin sulfate and incubated at 20°C for 24 h. After the incubation period the developing colonies were examined. *Saprolegnia* was characterized by an external, cotton-like appearance. Hyphae were subcultured in PDA media several times until a purified colony was obtained, which was then stored on PDA at 4°C. The morphology of the fungal mycelium was also studied during this isolation process.

### Fungal identification

**DNA extraction.** Plates with purified colonies of *Saprolegnia* were selected for DNA extraction. Fungal hyphae were placed aseptically into a 2 ml sterile centrifuge tube and washed successively with 0.8% sterile saline solution, EDTA and sterile dH<sub>2</sub>O, then dried on filter paper (Xin Xing brand, Hangzhou Special Paper Co., Ltd.). All hyphae were transferred to a 10 ml glass homogenizer and ground for 5 min on ice with 10 µl each of Tris-Hcl and EDTA; the solution was transferred to an eppendorf tube, with 500 µl dH<sub>2</sub>O added. The fungal DNA was extracted with TaKaRa Mini-BEST Universal Genomic DNA Extraction Kit Ver.5.0. Code number 9765.

### ITS and 26S rDNA D1/D2 amplification, sequencing and homology analysis

The ITS and 26S rDNA D1/D2 regions of the fungal hyphae were amplified using TaKaRa Fungi Identification PCR Kit, code number RR178. One microlitre of dH<sub>2</sub>O was used to replace template DNA as a negative control and the 1 µl of positive control DNA included in the kit was added. The PCR products were checked by agarose gel electrophoresis. Two distinct PCR-positive bands were chosen and cut from the gel then purified with the Gel DNA Purification Kit (TaKaRa). Sequencing of the bands was performed using the 3730 DNA sequencer of Shanghai Invitrogen Biotechnology Co. Ltd., Shanghai, China. The sequences were analyzed using DNA Star software and a GenBank BLAST search ([http://www.moa.gov.cn/zwlml/tzgg/gg/201104/t20110422\\_1976324.htm](http://www.moa.gov.cn/zwlml/tzgg/gg/201104/t20110422_1976324.htm)). The NCBI blast tool was used to compare the 26S rDNA D1/D2 and the ITS sequence with other sequences in the GenBank database.

### Plant material and extract preparation

Thirty traditional herbal plant species (Table 1) used in Chinese folk medicine were purchased from the Ya'an Medical Company and identified by L. Zhao at the Department of Pharmacy, SiChuan Agricultural University. The plants were placed in an oven at 40°C until completely dried, then crushed separately and reduced to a coarse powder followed by pulverizing through a 60 mesh disintegrator to obtain a finer powder then stored at -20°C until used. Dried herb powder (8 g) from each of the 30 plant species was placed separately in a thimble of a Soxhlet apparatus and extracted with 120 ml ethanol for 1.5 h in a 75°C water bath. This was repeated three times. The extracts were then filtered through paper (Xin Xing brand, Hangzhou Special Paper Co., Ltd.) and evaporated in a vacuum rotary evaporator at 120 rpm min<sup>-1</sup>, 0.09 Mpa at 75°C until the solution no longer flowed. The material was weighed and the container including the extract was placed in an oven to dry for approximately 24 h. The extracts were then dissolved with 95% ethanol to 200 mg ml<sup>-1</sup> (dosage/solvent); this stock solution was used to prepare the desired concentrations for testing antifungal activity.

### Antifungal effects of different extracts

A dish dilution method was modified from Hu et al. (2013) to evaluate the antifungal activity. Potato dextrose agar media, including a final working concentration of herbal extracts of 2 g L<sup>-1</sup>, was used based on preliminary studies. Each herbal plant species was tested in triplicate. A rapeseed microplate method was also used in the study. Whole rapeseeds were autoclaved at 121°C for 20 min. The sterilized rapeseeds were placed around the exact edge of the *Saprolegnia* colony, and the agar plates incubated at 20°C for a further 48 h. The *Saprolegnia*-infected rapeseeds that had visible hyphae were transplanted onto the prepared PDA plates containing the herbal plant extracts, and incubated at 20°C for an additional 48 h. Plates containing PDA, sterile water and infected rapeseeds were considered positive controls; plates containing sterile rapeseeds were considered negative controls. Formalin, malachite green and 95% ethanol treated rapeseeds were also included as chemical controls for the herbal plant extract. The results were recorded visually by determining mycelium presence as positive (+) or negative (-).

### Minimum inhibitory concentration (MIC) evaluation

Herbal plant extracts that showed negative for the presence of mycelia growth at 2 g L<sup>-1</sup> in the screening study were tested for MIC evaluation. The antifungal susceptibility and the MIC evaluation test were carried out using a modified method of the Clinical and Laboratory Standards Institute (CLSI) M38-A2 for broth dilution antifungal susceptibility testing of filamentous fungi (CLSI (Clinical and Laboratory Standards Institute), 2002). The minimum inhibitory concentration was defined as the lowest concentration of crude

Table 1  
Antifungal activity and minimum inhibitory concentration (MIC) evaluation tests for 30 Chinese herbal plant extracts at concentrations of 2 g L<sup>-1</sup>

| No. | Species                            | Common medical name        | Family                  | Tested parts                                                        | Production Place | The inhibition ability | MIC(mg L <sup>-1</sup> ) |
|-----|------------------------------------|----------------------------|-------------------------|---------------------------------------------------------------------|------------------|------------------------|--------------------------|
| 1   | <i>Xanthium sibiricum</i>          | Siberian Cock Fruit        | <i>Asteraceae</i>       | Fruit with Involucre                                                | GuangXi          | +                      |                          |
| 2   | <i>Arctium lappa</i>               | Great Burdock Achene       | <i>Asteraceae</i>       | Fruit                                                               | JiangShu         | +                      |                          |
| 3   | <i>Sophora tonkinensis</i>         | Vietnamese Sophora Root    | <i>Leguminosae</i>      | Root                                                                | GuangXi          | +                      |                          |
| 4   | <i>Litsea cubeba</i>               | Mountain Spicy Tree Fruit  | <i>Lauraceae</i>        | Fruit                                                               | SiChuan          | +                      |                          |
| 5   | <i>Rheum officinale</i>            | Rhubarb                    | <i>Polygonaceae</i>     | Root and Rhizome                                                    | SiChuan          | +                      |                          |
| 6   | <i>Atractylodes lancea</i>         | Atractylodes Rhizome       | <i>Asteraceae</i>       | Rhizome                                                             | ShanXi           | +                      |                          |
| 7   | <i>Punica granatum</i>             | Pomegranate Rind           | <i>Punicaceae</i>       | Peel                                                                | SiChuan          | +                      |                          |
| 8   | <i>Dictamnus dasycarpus</i>        | Densefruit Pittany Bark    | <i>Rutaceae</i>         | Root bark                                                           | AnHui            | +                      |                          |
| 9   | <i>Syzygium aromaticum</i>         | <i>Syzygium aromaticum</i> | <i>Myrtaceae</i>        | Bud                                                                 | ShanXi           | —                      | 500                      |
| 10  | <i>Psoralea corylifolia</i>        | Malaytea Scurfpea Fruit    | <i>Leguminosae</i>      | Fruit                                                               | SiChuan          | +                      |                          |
| 11  | <i>Melia toosendan</i>             | Szechwan Chinaberry Fruit  | <i>Lauraceae</i>        | Fruit                                                               | SiChuan          | +                      |                          |
| 12  | <i>Belamcanda chinensis</i>        | Blackberry lily Rhizome    | <i>Iridaceae</i>        | Rhizome                                                             | HuNan            | +                      |                          |
| 13  | <i>Coptis chinensis</i>            | Golden Thread              | <i>Ranunculaceae</i>    | Rhizome                                                             | ChongQing        | +                      |                          |
| 14  | <i>Magnolia officinalis</i>        | Officinal Magnolia Bark    | <i>Magnoliaceae</i>     | Root bark                                                           | ChongQing        | —                      | 62.5                     |
| 15  | <i>Melaphis chinensis</i>          | Chinese Gall               | <i>Anacardiaceae</i>    | Galls infestation by Chinese sumac aphids (Melaphis chinensis Bell) | SiChuan          | —                      | 250                      |
| 16  | <i>Angelica pubescens</i>          | Doubleteeth                | <i>Umbelliferae</i>     | Root                                                                | An Hui           | +                      |                          |
| 17  | <i>Taraxacum mongolicum</i>        | Pubescent Angelica Root    | <i>Asteraceae</i>       | All the grass                                                       | AnHui            | +                      |                          |
| 18  | <i>Euphorbia fischeriana Steud</i> | Mongolian Dandelion Herb   | <i>Euphorbiaceae</i>    | Root                                                                | SiChuan          | —                      | 62.5                     |
| 19  | <i>Pseudolarix kaempferi</i>       | Langdu Root                | <i>Pinaceae Lindl.</i>  | Root bark                                                           | AnHui            | +                      |                          |
| 20  | <i>Notopterygium incisum</i>       | Incised Notopterygium      | <i>Umbelliferae</i>     | Rhizome and Root                                                    | SiChuan          | +                      |                          |
| 21  | <i>Cnidium monnieri</i>            | Common Cnidium Fruit       | <i>Umbelliferae</i>     | Fruit                                                               | GuangXi          | +                      |                          |
| 22  | <i>Sophora flavescens</i>          | Lightyellow Sophora Root   | <i>Leguminosae</i>      | Root                                                                | GuangXi          | —                      | 250                      |
| 23  | <i>Paeonia veitchii</i>            | Red Peony Root             | <i>Ranunculaceae</i>    | Root                                                                | SiChuan          | +                      |                          |
| 24  | <i>Gardenia jasminoides</i>        | Cape Jasmine Fruit         | <i>Rubiaceae</i>        | Fruit                                                               | JiangXi          | +                      |                          |
| 25  | <i>Aristolochia debilis</i>        | Dutchmanspipe Fruit        | <i>Aristolochiaceae</i> | Fruit                                                               | Shan Dong        | +                      |                          |
| 26  | <i>Quisqualis indica</i>           | Rangooncreeper Fruit       | <i>Combretaceae</i>     | Fruit                                                               | SiChuan          | +                      |                          |
| 27  | <i>Peucedanum praeruptorum</i>     | Whiteflower Hogfennel      | <i>Umbelliferae</i>     | Root                                                                | AnHui            | +                      |                          |
| 28  | <i>Saposhnikovia divaricata</i>    | Root                       | <i>Umbelliferae</i>     | Root                                                                | HeBei            | +                      |                          |
| 29  | <i>Scutellaria baicalensis</i>     | Divaricate Saposhnikovia   | <i>Labiateae</i>        | Root                                                                | JiLin            | +                      |                          |
| 30  | <i>Prunus mume</i>                 | Baical Skullcap Root       | <i>Roseceae</i>         | Nearly ripe fruit                                                   | FuJian           | +                      |                          |
| 31  | 95% ethanol                        | Dark Plum Fruit            |                         |                                                                     |                  |                        | 1000                     |
| 32  | Malachite green                    |                            |                         |                                                                     |                  |                        | 1.25                     |
| 33  | Formalin                           |                            |                         |                                                                     |                  |                        | 2.76                     |

Only herbal plant extracts negative for hyphal growth on rapeseeds at 2 g L<sup>-1</sup> in the screening study were tested for MIC evaluation. Five herbs *Syzygium aromaticum*, *Magnolia officinalis*, *Melaphis chinensis*, *Euphorbia fischeriana Steud*, *Sophora flavescens* exhibited growth inhibition at this concentration.

Note: '+', fungal hyphae grew on the media with medicine; '—', no fungal hyphae growth.

extract that prevented visible growth or germination of spores.

#### Spore preparation

Thirty infected rapeseeds were subcultured in 200 ml sterile pond water collected from the SAU teaching fish farm pond and autoclaved before use with 200 mg L<sup>-1</sup> streptomycin sulphate at 20°C until encystment. A Vortex mixer was used to agitate the subculture for 10 min to release the spores. The

solution was filtered with three layers of sterile gauze. The final spore concentration was calculated using a haemocytometer adjusted to 1.0 × 10<sup>4</sup> CFU ml<sup>-1</sup> with sterile water.

#### Broth preparation

Ninety-six well plates were used in evaluating MIC of different extracts. All extracts were prepared in serial two-fold dilutions in RPMI-1640 broth within a dilution range from 1 : 1 to 1 : 512. Each well contained 50 µl grade of diluted



extracts and 50  $\mu$ l spore suspension. In addition, one well acted as the positive control containing only the herbal extract; another well contained only the spore suspension and acted as the negative control. The plates were incubated at 20°C for 48 h. Each herbal extract was tested in triplicate.

## Results

### Fungal morphology

The CCF1301 strain of *Saprolegnia* used in this study was isolated from the tail skin of infected grass carp (*Ctenopharyngodon idella*). The hyphae were transparent, tubular, non-septa, and sparingly branched (Fig. 1a). The sporangiums were clavate, fusiform, straight or bent and the spore arranged in several lines. Mature spores were released from the sporangiums and swam away quickly (Fig. 1b). Archeogonia were spherical and appeared in the same hyphae or different hyphae with antheridium. This morphology was similar to earlier research published on *Saprolegnia* sp. (Stuehl et al., 2005; Ke et al., 2009).

### DNA sequencing and homology analysis

Two fragments, about 750 bp, of 26S rDNA D1/D2 and ITS region of CCF1301 were amplified. The sequencing results showed that the two genes were 839 and 798 bp. These gene sequences were submitted to NCBI and subsequently assigned NCBI accession numbers KJ577573 and KF871196, respectively. Using the blast tool of the NCBI to compare 26S rDNA D1/D2 sequence with other sequences in the GenBank database, the total score between CCF1301 and *Saprolegnia* sp. JN400036.1 was highest, with a similarity rate of 99% and query coverage reaching 88%. Furthermore, using the online blast tool of the NCBI, the total score between CCF1301 ITS sequence and *Saprolegnia ferax* JN400035.1 was the highest, with the similarity rate and query coverage attaining 99 and 96%, respectively. From the blasting results of the two genes, it was concluded that CCF1301 could be identified as *Saprolegnia ferax*.

### Antifungal activities of different extracts

The antifungal activities of the thirty herbal extracts are shown in Table 1. Among all herbal extracts, *Syzygium aromaticum*, *Magnolia officinalis*, *Melaphis chinensis*, *Euphorbia fischeriana* Steud, and *Sophora flavescens* exhibited the best inhibition of growth. MIC values of the herbal extracts were 500, 62.5, 250, 62.5, 250 mg L<sup>-1</sup>, respectively. At the same time, MIC values for 95% ethanol, malachite green and formalin were 1000  $\mu$ l L<sup>-1</sup>, 1.25 mg L<sup>-1</sup> and 2.76  $\mu$ l L<sup>-1</sup>, respectively. From the results we conclude that *Magnolia officinalis* and *Euphorbia fischeriana* Steud had the best antifungal activity and could be used for further research.

## Discussion

Saprolegniosis is a common disease occurring in many species of fish worldwide (Shahbazian et al., 2010; Sarowara

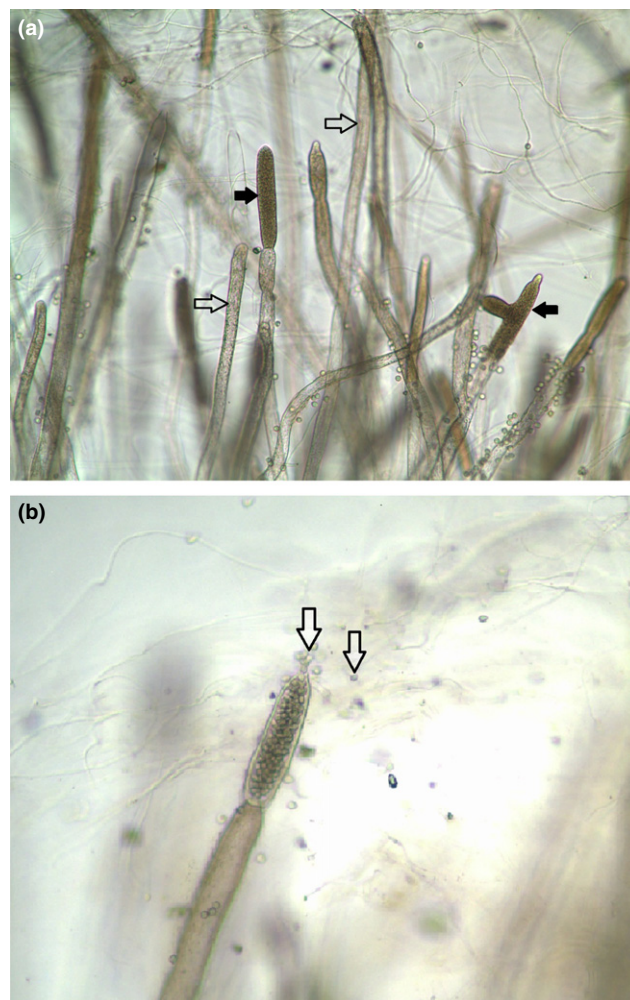


Fig. 1. Morphology of hyphae and zoosporangia of *Saprolegnia* strain CCF1301 (subsequently identified as *S. ferax*) isolated from cultured grass carp (*Ctenopharyngodon idella*) with gross signs of saprolegniosis. Hyphae are cultured on potato dextrose agar (PDA) plates with 200 mg L<sup>-1</sup> streptomycin sulfate and incubated at 20°C for 24 h. (a) Hyphae transparent tubular, non-septa, sparingly branched. Mature zoosporangia clavate or fusiform, straight or bent or furcal and usually separated by a thin membrane. Horizontal arrow = hyphae; black arrow = zoosporangia. (b) Mature spores being released from the zoosporangia apex. Vertical arrow = mature spores

et al., 2013) and often causes serious damage, especially to incubating eggs. Malachite green is an effective fungicide, especially as a general fish hatchery disinfectant; while approved for use in aquarium fish, this is no longer the case for use in fish intended for human consumption (van West, 2006; Sudova et al., 2007). With restrictions on the use of malachite green, more emphasis has been directed toward identifying suitable alternatives. Alternative antifungal agents have focused mainly on three candidates: formalin (Barnes and Soupir, 2006; Akpoiloh and Adebayo, 2010), hydrogen peroxide (Rach et al., 2004; Mitchell et al., 2009; Matthews et al., 2012), and sodium chloride or seawater (Waterstrat and Marking, 1995; Koepudsa et al., 2005; Ali, 2009). Formalin is widely used as a topical treatment for

saprolegniosis; however, it has the potential of being toxic, particularly at temperatures above 25 °C (Akpoilih and Adebayo, 2010) and reported dosages of 50–150 mg L<sup>-1</sup> (Barnes and Soupir, 2006; Akpoilih and Adebayo, 2010), and is impractical for use in large ponds. In addition, formalin also poses a risk to human health. Other compounds have also been investigated, for example, sodium hypochlorite (NaOCl) was thought to be a remarkable and effective antifungal chemical to prevent fungal infection on the eggs of white-spotted char, *Salvelinus leucomaenis*, rainbow trout *Oncorhynchus mykiss* and chum salmon *O. keta*, but Kashiwagi et al. (2007) found that it could lead to egg softening at certain concentrations. Good efficacy on control of *S. parasitica* infection in grass carps was observed with copper sulfate (Sun et al., 2014); Cao et al. (2012) also found that a concentration of dithiocyano-methane, another compound used to treat saprolegniosis, can influence the zoospore number of *Saprolegnia ferax*. None of these therapeutants, however, overcome the issue of applying treatments to large ponds. In China, traditional medicines have been used as treatments for disease in pond cultured fish for millennia (Wu and Li, 2009). It is proposed that purification and synthesis of the principal antifungal components of the herbal extracts could be of considerable value because of their abundance, low cost and low residues. These would be most practical for use in either feed or bath, using a synthetic chemical after a clear determination of the principal antifungal components of the herbal extracts.

Chinese herbs comprise thousands of species with a multitude of functions. However, a single species may also contain many different constituents and as such their activity can be quite diverse and multitudinous. Herbal extracts are also often used in combinations, whereby the principal ingredient provides the main therapeutic thrust, and the second principal ingredient may enhance or assist the therapeutic actions of the first. Any additional ingredient would: treat accompanying symptoms, moderate the harshness or toxicity of the primary symptoms, guide the medicine to the proper organs, or exert a harmonizing effect (Yuan and Lin, 2000). While it is possible to separate and determine the pharmacological function of each ingredient in each herb, because of the multitude of Chinese herbs a screening procedure must first be employed to identify the specific efficacy of the compound(s) in the microorganisms being studied. In the present study, we identified thirty herbs, based on reported antifungal activity following a preliminary inspection of the literature, for potential efficacy against *Saprolegnia ferax*. Included in this group are nineteen novel herbal extracts. Five herbal extracts with improved antifungal activity were identified, including *Syzygium aromaticum*, *Magnolia officinalis*, *Melaphis chinensis*, *Euphorbia fischeriana* Steud, and *Sophora flavescens*. This is the first report of antifungal, and indeed, anti-*Saprolegnia* activity in *Euphorbia fischeriana* Steud and *Sophora flavescens*. Of these five herbal extracts, *Magnolia officinalis* and *Euphorbia fischeriana* Steud were found to have the most effective antifungal activity. *Magnolia officinalis* has been reported to have several functional properties including antioxidation, antiviral, antineoplastic, antibiosis and analgesia properties (Fang et al., 2009; Zhang et al., 2013a; Zhang

et al., 2013b). Jin (2011) demonstrated the main ingredients of *Magnolia officinalis* to be magnolol and honokiol, which showed good inhibition of *Candida albicans*. But whether these two ingredients had antifungal activity in *Saprolegnia* sp. still needs further experimentation for confirmation. *Euphorbia fischeriana* Steud has been shown to have a good efficacy in treating fungal diseases of the skin, tuberculosis and certain cancers in humans (Fang, 2002; Zhang et al., 2013a,b). Some commercially available topical ointments include this herbal extract as a treatment for fungal diseases in humans. This is the first report of *Euphorbia fischeriana* Steud demonstrating good antifungal activity against *Saprolegnia*. However, it should be noted that *Euphorbia fischeriana* Steud has a high natural toxicity (Yang, 2012) and therefore its applicability as the main ingredient in a therapeutic aquaculture formulation would require further research. *Arctium lappa* and *Cnidium monnieri* were reported by Hu et al. (2013) to have useful antifungal activity; however, our results and those of Cao et al. (2012) noted that *Cnidium monnieri* did not inhibit *Saprolegnia* sp. growth. It is possible that this discrepancy might be a reflection of the different geographical regions and within-species variation of the herbal extracts and *Saprolegnia* sp. For clarity, Table 1 includes the production location for each of the herbal species used in this study.

From this study we conclude that *Magnolia officinalis* and *Euphorbia fischeriana* Steud had the most effective antifungal activity against *Saprolegnia* sp. These preliminary screening results are very promising. Our future plans are to continue to identify the specific active compounds in *Magnolia officinalis* and *Euphorbia fischeriana* Steud using systemic solvent methods and to evaluate the antifungal capability of the different chemical fractions to find the most effective solvent. A chromatographic methodology should also assist in separating the individual monomers to identify those most effective against *Saprolegnia* sp. It is hoped this will eventually lead to the development of an effective, novel and natural treatment against saprolegniosis.

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